

MATHEMATICAL EXPECTATION

MEANING OF MATHEMATICAL EXPECTATION

Historically, the term *mathematical expectation* or *expected value* is derived from games of chances. In such games, the gamblers were concerned with how much, on the average, one would expect to win if the game is continued for a sufficiently long time. In statistical terminology, the *expected value* or *mathematical expectation* of a random variable is associated with a sample space and is, in fact, the average value of the random variable generated through a random experiment. As you will see, the expected value of a random variable is closely related to the notion of the arithmetic mean. It is generally known as the *mean of the population*.

Definition:

If X is a discrete random variable defined over the sample space $\{x_1, x_2, \dots, x_n\}$ with corresponding probabilities $P(x_1), P(x_2), \dots, P(x_n)$, then the expected value or the mathematical expectation of X is symbolically defined as

$$E(X) = \sum_{i=1}^n x_i P(x_i)$$

If X is continuous having a density function $f(x)$, then

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

The $E(X)$ is thus the mean value of X and the term expectation, expected value, mathematical expectation, or population mean can be used interchangeably.

Properties of Expectation

- | | |
|---------------------------------------|------------------------------|
| 1. Linearity: | $E[aX + bY] = aE[X] + bE[Y]$ |
| 2. Expectation of a constant: | $E[c] = c$ |
| 3. Multiplication by constant: | $E[aX] = a$ |
| 4. If X and Y are independent: | $E[XY] = E[X] \cdot E[Y]$ |

Examples

 **Example 1: Rolling a fair die**

Let X be the number on the die:

$$E[X] = \sum_{i=1}^6 i \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Example 2: Tossing a coin

Let $X = 1$ for heads, $X = 0$ for tails.
If the coin is fair:

$$E[X] = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}$$

Example: Find the expected value of the random variable X having the following density function:

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Solution: By definition

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = 2 \int_0^1 x(1-x)dx \\ &= 2 \int_0^1 xdx - 2 \int_0^1 x^2 dx \\ &= (x^2)_0^1 - \frac{2}{3}(x^3)_0^1 \\ &= 1 - \frac{2}{3} \\ &= \frac{1}{3} \end{aligned}$$

EXPECTED VALUE OF A FUNCTION OF A RANDOM VARIABLE

If X is a random variable, then X^2 , $2X$, $3X + 1$, $(X - 1)^2$, $2a + X$, are all functions of X and Y definition, they are also random variables. Each of these random variables has a probability distribution identical to that of X and thus has an expected value, which can be computed in usual manner.

Example: Let the random variable X have the following probability distribution:

x	-1	0	1
$f(x)$	0.2	0.3	0.5

Compute the expected value of (i) $E(X)$ (ii) $E(2X)$ (iii) $E(3X + 1)$

Solution:

$$(i) \quad E(X) = \sum xf(x) = -1 \cdot 0.2 + 0 \cdot 0.3 + 1 \cdot 0.5 = 0.3$$

(ii) To compute $E(2X)$, we prepare the following table:

Values of $2x$	-2	0	2
$f(x)$	0.2	0.3	0.5

Since, $2X = -2$ is equivalent to $X = -1$, the computation of $P(2X = -2)$ will be equivalent to the computation of $P(X = -1)$. Hence, $P(2X = -2) = P(X = -1) = f(-1) = 0.2$

Similarly, $P(2X = 0) = P(X = 0) = f(0) = 0.3$ and

$P(2X = 2) = P(X = 1) = f(1) = 0.5$.

Now,

$$E(2X) = (-2)(0.2) + 0(0.3) + 2(0.5) = -0.4 + 0 + 1 = 0.6$$

(iii) To compute $E(3X+1)$, we prepare the following table:

Values of $3x+1$	-2	1	4
$f(x)$	0.2	0.3	0.5

$$E(3X+1) = 3E(X) + 1 = 3(0.3) + 1 = 0.9 + 1 = 1.9$$

EXPECTED VALUE OF A FUNCTION OF TWO RANDOM VARIABLES

The notion of mathematical expectation can be extended to functions of two or more random variables. We will deal here with the case of two variables, which can analogously be extended for 3 or more variables. Let X and Y be two random variables with joint probability distribution $f(x, y)$. The expected value of the function $w(X, Y)$ is defined as

$$E[w(X, Y)] = \sum_x \sum_y w(x, y) f(x, y), \quad \text{if } X \text{ and } Y \text{ are discrete, and}$$

$$E[w(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} w(x, y) f(x, y) dx dy, \quad \text{if } X \text{ and } Y \text{ are continuous.}$$

Example:

Given the following joint density function of X and Y :

$$f(x, y) = 4xy, \quad 0 < x < 1, 0 < y < 1$$

$$= 0, \quad \text{elsewhere.}$$

Obtain $E(X)$, $E(Y)$, $E(XY)$, and $E(X + Y)$.

Solution:

$$\begin{aligned} \text{i. } E(XY) &= \int_0^1 \int_0^1 xy f(x, y) dx dy \\ &= 4 \int_0^1 \int_0^1 x^2 y^2 dx dy \end{aligned}$$

$$\begin{aligned}
&= \frac{4}{3} \int_0^1 (x^2 y^3)_0^1 dx \\
&= \frac{4}{3} \int_0^1 x^2 dx \\
&= \frac{4}{9} (x^3)_0^1 \\
&= \frac{4}{9}
\end{aligned}$$

ii. $E(X) = \int_0^1 \int_0^1 x f(x, y) dx dy$

$$\begin{aligned}
&= 4 \int_0^1 \int_0^1 x^2 y dx dy \\
&= \frac{4}{2} \int_0^1 (x^2 y^2)_0^1 dx \\
&= \frac{4}{2} \int_0^1 x^2 dx \\
&= \frac{4}{6} (x^3)_0^1 \\
&= \frac{2}{3}
\end{aligned}$$

iii. $E(Y) = \int_0^1 \int_0^1 y f(x, y) dx dy$

$$\begin{aligned}
&= 4 \int_0^1 \int_0^1 x y^2 dx dy \\
&= \frac{4}{3} \int_0^1 (x y^3)_0^1 dx \\
&= \frac{4}{3} \int_0^1 x dx \\
&= \frac{4}{6} (x^2)_0^1 \\
&= \frac{2}{3}
\end{aligned}$$

iv. $E(X + Y) = \int_0^1 \int_0^1 (x + y) f(x, y) dx dy$

$$\begin{aligned}
&= 4 \int_0^1 \int_0^1 (x^2 y + x y^2) dx dy \\
&= 4 \int_0^1 \left(\frac{1}{2} x^2 y^2 + \frac{1}{3} x y^3 \right)_0^1 dx \\
&= 4 \int_0^1 \left(\frac{1}{2} x^2 + \frac{1}{3} x \right) dx
\end{aligned}$$

$$= 4\left(\frac{1}{6}x^3 + \frac{1}{6}x^2\right)_0^1$$

$$= \frac{4}{3}$$

Example: Given the following density function

$$f(x, y) = 2(x + y - 2xy), \quad 0 < x < 1, 0 < y < 1$$

$$= 0, \quad \text{elsewhere}$$

- (a) Find $E(X)$, $E(Y)$, $E(X + Y)$ and $E(XY)$.
 (b) Also verify whether $E(X + Y) = E(X) + E(Y)$.
 (c) Are X and Y independent?

Solution:

The marginal density of X is $g(x) = 2 \int_0^1 (x + y - 2xy) dy$

$$= 2\left(xy + \frac{y^2}{2} - \frac{2xy^2}{2}\right)_0^1$$

$$= 2\left(x + \frac{1}{2} - x\right)$$

$$= 1$$

Hence,

$$E(X) = \int_0^1 xg(x)dx$$

$$= \int_0^1 xdx$$

$$= \frac{1}{2}$$

Similarly

$$h(y) = 2 \int_0^1 (x + y - 2xy) dx$$

$$= 2\left(\frac{x^2}{2} + xy - \frac{2yx^2}{2}\right)_0^1$$

$$= 2\left(\frac{1}{2} + y - y\right)$$

$$= 1$$

And

$$E(Y) = \int_0^1 yh(y)dy$$

$$= \int_0^1 ydy$$

$$= \frac{1}{2}$$

Now,

$$\begin{aligned}
 E(X + Y) &= 2 \int_0^1 \int_0^1 (x + y)(x + y - 2xy) dx dy \\
 &= 2 \int_0^1 \int_0^1 (x^2 + y^2 + 2xy - 2x^2y - 2xy^2) dx dy \\
 &= 2 \int_0^1 \left(x^2y + \frac{y^3}{3} + xy^2 - x^2y^2 - \frac{2xy^3}{3} \right) \Big|_0^1 dx \\
 &= 2 \int_0^1 \left(x^2 + \frac{1}{3} + x - x^2 - \frac{2x}{3} \right) dx \\
 &= 2 \int_0^1 \left(\frac{1}{3} + \frac{x}{3} \right) dx \\
 &= \frac{2}{3} \int_0^1 (1 + x) dx \\
 &= \frac{2}{3} \left(x + \frac{x^2}{2} \right) \Big|_0^1 \\
 &= 1
 \end{aligned}$$

Since $E(X) + E(Y) = \frac{1}{2} + \frac{1}{2} = 1$, we have $E(X) + E(Y) = E(X + Y)$.

$$\begin{aligned}
 E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy \\
 &= 2 \int_0^1 \int_0^1 xy(x + y - 2xy) dx dy \\
 &= 2 \int_0^1 \int_0^1 (x^2y + xy^2 - 2x^2y^2) dx dy \\
 &= 2 \int_0^1 \left(\frac{x^3y}{3} + \frac{x^2y^2}{2} - \frac{2x^3y^2}{3} \right) \Big|_0^1 dy \\
 &= 2 \int_0^1 \left(\frac{y}{3} + \frac{y^2}{2} - \frac{2y^2}{3} \right) dy \\
 &= 2 \left(\frac{y^2}{6} + \frac{y^3}{6} - \frac{2y^3}{9} \right) \Big|_0^1 \\
 &= \frac{2}{9}
 \end{aligned}$$

Now

$$E(X)E(Y) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \text{ and } E(XY) = \frac{2}{9}$$

Since $E(X)E(Y) \neq E(XY)$, the variables are not independent.

Exercise:

1. Given the following joint density function of the random variables X and Y.

$$f(x, y) = \frac{x(3y^2 + 1)}{4}, \quad 0 < x < 2, 0 < y < 1$$
$$= 0, \quad \text{elsewhere}$$

- a) Examine whether X and Y are independent.
b) Also check that $E(X + Y) = E(X) + E(Y)$.
c) Find $E\left(\frac{Y}{X}\right)$